

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with **6+ years** building production **Node.js** services, message-driven pipelines and data migrations that run under live traffic. Led a **zero-downtime migration of 20+ production applications** at MTailor on a real-time two-way sync built with **Google Pub/Sub** — idempotent consumers, redelivery assumed by default, every service individually rollback-able throughout the cutover. Also builds **LLM systems that have to be right**: an agentic research pipeline that searches the open web, fetches and verifies each source, scores it and discards the rest, and a production WhatsApp assistant on OpenAI + Google Vision for prescription OCR. Comfortable across the stack — React front ends in three roles — and blunt about cost: cut **\$5,000/month** of recurring cloud spend by consolidating the estate.

CORE COMPETENCIES

Node.js Backends	LLM Pipelines & Retrieval	Queues, Retries & Idempotency	Event-Driven Architecture
Zero-Downtime Migrations	SQL & NoSQL Data Modelling	Ranking & Recommendation	Cost Optimisation / FinOps
			React

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Built **real-time two-way sync** between MongoDB and Firestore on **Node.js + Google Pub/Sub** — idempotent consumers designed for redelivery, keeping both datastores serving live traffic through the entire cutover window.
- Cut recurring infrastructure spend by **\$5,000/month** by consolidating all services off AWS onto GCP; migrated file storage from S3 to GCS via Python.
- Shipped **Express Shipping** end to end — backend, pricing and ops handoff — generating **+\$40 revenue per order**; built a 3D visualisation feature (ffmpeg video overlay) that improved buyer conversion.

MTailor

Mar 2022 – Jul 2022

Backend Contractor · US Remote

- Built the MongoDB→Firestore migration pipeline, applying complex processing logic on the way through — on a stack that was new to him at the start of the contract.
- Wrote the Firebase SDK documentation (Node.js and Python) that other engineers onboarded from, and trained the Ops team on dashboards and backup procedures. Converted to full-time and given the whole migration programme.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimisation of a PWA serving **real-time stock prices**; built scheduled jobs querying **2M+ Firestore records** to keep customer-facing prices current from Morningstar APIs.
- Extended backend APIs with Firebase Cloud Functions and **Express.js**; added responsive React/Tailwind pages to the same product.

Mind2matter

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built **DeFi backends** on Node.js and Web3 for a US agency — money-moving paths where a mistake is irreversible — reporting directly to the CTO; delivered React UIs from Figma under tight timelines.

CodeBits

Jan 2020 – Jul 2021

Software Engineer / Team Lead · Uganda Remote

- Architected an **event-driven microservices backend** for FIDA Uganda's case-management platform using **Apache Kafka, Docker and Kubernetes**; led a team of 4 developers.
- Built a USSD legal-aid service talking to a Node.js backend over **gRPC**, with a provider-agnostic contract multiple providers integrated against without bespoke code.
- Shipped a paralegal database and a cross-platform React Native/Expo mobile app with Firebase push notifications.

PROJECTS

Agentic Research & Grading Pipeline

Searches job portals and forums, fetches each posting, verifies the canonical source in a headless browser before anything is scored, grades the result against a profile and discards the rest — 200+ evaluations. Built after learning that aggregator metadata routinely lies, so the verification gate sits *before* the scorer.

Node.js · LLM APIs · Playwright · retrieval & scoring · dedup

Healthcare WhatsApp Assistant with OCR

Production conversational assistant on OpenAI with Google Vision prescription OCR behind the WhatsApp Business API — a correctness-critical path where dialogue state, media ingestion and platform limits shaped the architecture as much as the model.

Node.js · OpenAI · Google Vision · WhatsApp Business API

Marketplace Ranking & Recommendation Engine

Collaborative and content-based filtering over a messy, duplicate-prone provider catalog, emitting ranked confidence rather than binary match/no-match so the ambiguous middle stays reviewable.

Node.js · MongoDB

Real-time Pub/Sub Sync Pipeline

Live two-way MongoDB↔Firestore sync running under parallel production traffic for the duration of a zero-downtime migration, with idempotent consumers and conflict handling.

Node.js · Google Pub/Sub · MongoDB · Firestore

EDUCATION

B.Sc. Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

LINKS

Open-source & project code — github.com/zac-09

SKILLS

Backend: Node.js, Express.js, REST APIs, gRPC, Microservices, Event-Driven Design

Messaging & Data: Google Pub/Sub, Apache Kafka, NATS Streaming, MongoDB, Firestore, SQL

AI / LLM: OpenAI APIs, Google Vision OCR, retrieval & ranking, agentic pipelines

Cloud / Infra: GCP, Firebase (Cloud Functions), AWS, Docker, Kubernetes, Linux

Languages: JavaScript, TypeScript, Python, Go (intermediate), Rust (intermediate) **Frontend:** React, React Native, Tailwind, HTML, CSS

Spoken: English, Kiswahili